

Scheme 1

Stage 1 Screening and Eligibility Decision Scheme

Title and abstract eligibility for the BPS chronic pain corpus

DRAFT FOR EXPERT EVALUATION. These coding schemes are a working draft circulated for expert evaluation. They have not been applied to a final review corpus. The current manuscript is a test run that exercised an earlier, coarser generation of these schemes. The workflow itself has since been validated end to end in two cross-provider test runs, in which three large language models from three different providers applied the abstract-level and the full-text scheme independently and their agreement was quantified per coded field. The full run on the review corpus is deliberately held until this evaluation is complete.

At a Glance

Workflow position

Pre-extraction eligibility screen, run after search and deduplication and before Stage 2 abstract coding.

Operational mode

Deterministic rule set that emits a provisional decision, confidence, and reason. Every decision is validated by a human screener in Rayyan.

Unit of analysis

One bibliographic record (title, abstract, and publication metadata).

Provenance basis

Executable screening rules plus the OSF eligibility criteria.

Research questions

Gatekeeper for RQ1, RQ2, RQ3 (defines the analysable corpus)

Tagline

Rule-based provisional machine assist with mandatory human validation

Canonical Source Paths

- `src/01_protocol/decision_rules/screening_rules.md`
- `src/03_pipeline/bps_review/screening/rules.py`
- `src/09_review_stages/03_screening/README.md`
- `src/01_protocol/osf/OSF_registration-HTBMFCPR.md`

Purpose

This scheme operationalizes title and abstract screening after search and deduplication. Its function is to decide whether a record enters the review corpus for downstream coding, using a human-validatable rule set centred on biopsychosocial language, chronic pain relevance, review design, and population eligibility.

Because everything downstream inherits this decision, the scheme is deliberately conservative: it protects recall at the boundary and pushes genuinely ambiguous records into a borderline register rather than excluding them early.

Decision Fields and Controlled Values

The scheme writes one decision bundle per record. The controlled values below are the operational vocabulary.

stage1_decision

Eligibility verdict for the record.

- **include** Meets every inclusion rule with no triggered exclusion. *Choose when:* Review design, BPS term, chronic pain, adult or mixed population, English, in window. *Not this if:* Any hard exclusion trigger present. *Boundary:* Any single hard exclusion overrides inclusion.
- **exclude** At least one hard exclusion rule fires with high confidence. *Choose when:* Clear primary study, wrong population, non-English. *Not this if:* Only a soft or ambiguous signal, which is maybe. *Boundary:* Use exclude only when the trigger is unambiguous; otherwise use maybe.
- **maybe** Borderline record retained for human adjudication. *Choose when:* Ambiguous review type, mixed acute and chronic, unclear age or duration. *Not this if:* A clearly satisfied or clearly failed rule. *Boundary:* maybe is the honest label for genuine uncertainty and must not be used to avoid a decision that the abstract actually supports.
- **unclear** Current executable fallback state for ambiguous cases where maybe is not emitted. *Choose when:* Record cannot be resolved from available metadata. *Not this if:* A confident include or exclude. *Boundary:* unclear is the state the executable rule set emits when a record cannot be resolved at all.

stage1_reason

Controlled reason attached to exclusions and unclear cases. See the exclusion catalogue below.

stage1_confidence

Screener or rule confidence in the decision.

- **high** Decision rests on explicit, unambiguous metadata (for example an explicit non-review publication type).
- **medium** Decision rests on strong but partly inferential signals.
- **low** Decision rests on weak or conflicting signals; always pair with a logged rationale.

stage1_screened_by

Provenance of the provisional decision. Current outputs record `codex_machine_assist` before human validation.

stage1_screening_mode

Screening mode. Current outputs record `rule_based_provisional`.

Inclusion Logic

All of the following must hold for an include.

Review design

The record is a review article or review-like evidence synthesis (systematic, meta-analysis, scoping, umbrella, narrative, realist, integrative, or expert review).

BPS lexical trigger

The title or abstract explicitly contains a biopsychosocial term: biopsychosocial, bio-psycho-social, or bio psycho social.

Chronic pain focus

The focus concerns chronic pain, persistent pain, or a named chronic pain condition.

Population

The population is adult or mixed-age rather than pediatric-only.

Window and language

The record is within the operational search window and in English.

Exclusion Catalogue and Implemented Reason Labels

Each exclusion reason below is a controlled string with an operational trigger.

no biopsychosocial term in title/abstract

No explicit biopsychosocial term is present in the title or abstract.

outside operational search window

Publication date falls before or after the operational search dates configured in config/protocol.yaml.

protocol

Protocol papers without results are excluded.

commentary/editorial/letter

Commentary, editorial, and letter publication types are excluded.

animal/non-human focus

Animal-only or non-human records are excluded unless a human focus is explicit.

pediatric-only focus

Populations restricted to under 18 are excluded.

acute pain focus

Acute-only pain records are excluded when chronicity is not also explicit.

chronic pain focus unclear

Pain is present but chronic pain relevance is insufficiently clear.

non-English record

Records not in English are excluded.

review status unclear

The record cannot be confidently identified as a review or evidence synthesis from title, abstract, or publication metadata.

Borderline Handling

Ambiguous review type, mixed acute and chronic populations, unclear age group, or unclear pain duration should not be over-excluded. Borderline cases are logged with a rationale and lower confidence.

Musculoskeletal ambiguity is intentionally carried forward. Stage 3 full-text adjudication resolves it rather than Stage 1 excluding too aggressively.

Primary Outputs Using This Scheme

- `src/09_review_stages/03_screening/outputs/stage1_screening.csv`
- `src/09_review_stages/03_screening/audit/stage1_screening_summary.csv`
- `src/09_review_stages/03_screening/audit/reliability_report.csv`